DR5b: Where the Wall Shows Up — Verification Corollaries

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Abstract

DR5 established a structural limit: a proposition-ranking nominator N cannot distinguish a commitment D from any specific realisation r_i when D admits more than one non-equivalent surface form; escape requires importing an external grouping function g; and when g is inherited from a matcher, the projection-vs-genuine-signal ambiguity is inherited too. The DCR arc supplied ground truth (Einstein's 1905 deletion) for exactly one such D (absolute simultaneity) in exactly one corpus (pre-1905 electrodynamics).

DR5b enumerates four concrete verification settings outside the DCR arc where the theorem operates, with worked examples of the wall in each. It also gives specific guidance for verifier designers who want to check whether their protocol is above or below the DR5 wall for their specific target.

The claim is not that verification is impossible for multi-realisation targets. It is that the standard placebo-based test — "*our checker fires on this positive case; it does not fire on this control; therefore our checker works*" — is *insufficient* whenever the target admits multiple non-equivalent surface forms. Something stronger is required, and DR5 says exactly what.

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1. Recap: what DR5 says, in one paragraph

Let D be a target commitment with realisations $\{r_1, \dots, r_k\} \subseteq C$ in a corpus C. Let $N: C \to \mathbb{R}$ be a proposition-ranking scoring function: $N(p)$ depends only on p, not on any subset structure over C. Then N produces k distinct scores $\{N(r_i)\}$ that cannot be aggregated into a single "score for D" without importing an external grouping function g that identifies the realisation set. Once g is imported, the ranking is class-based, not proposition-based, and the correctness of g (soundness + completeness for D) becomes the load-bearing question.

The wall shows up as follows: a matcher tuned to catch r_i misses r_j ($j \neq i$); a matcher wide enough to catch all r_j overfires on the placebo; there is no calibration that resolves both without an external g.

2. Corollary 1 — Code correctness

Setting. A verifier checks whether an implementation satisfies a specification. The specification D is what the code should do; the implementation p is a specific source-code artifact. A proposition-ranking verifier scores p against D independently of other implementations.

Where D becomes multi-realisation.

Consider D = "*the code assumes 0-based indexing throughout.*" Surface realisations of this commitment in Python include:

- for i in range(len(xs)): ...
- for i, x in enumerate(xs): ... use i ...
- xs[0] as the first element
- Slicing xs[i:i+n] with the convention i starts at 0

- Comments like "# 0-indexed"

An implementation might embody D in one of these forms; a different implementation of the same functional spec might embody it in another. A verifier prompted with "*does this code assume 0-based indexing?*" without an explicit grouping function will either:

miss the others; or

code that happens to have xs[0] in a different logical position.

- **Match only one surface form** — e.g., only firing on xs[0] — and

- **Match all forms** with a wider pattern that also fires on 1-based

Where the placebo fails. A placebo (1-based indexing) can look identical at the surface level in short snippets. Verifiers that check by lexical matching (xs[0] appears) will fire on placebos where xs[0] is an aliased sentinel value rather than a first-element access. DR5 §5 says this is not a bug in the verifier — it is the wall.

Practical guidance. If the code-correctness target admits multiple surface realisations, do not rely on placebo-based verification of a proposition-independent scorer. Either supply an explicit g (a specification-derived AST rewrite that canonicalises all realisations to one form before scoring), or accept that the verifier operates *above* the DR5 wall only when the target has a canonical form.

3. Corollary 2 — LLM reasoning verification

Setting. An oracle checks whether a language model's reasoning trace supports its conclusion. The oracle scores each step independently. D is "*the model's answer is derived from valid reasoning steps.*"

Where D becomes multi-realisation. Valid reasoning has many surface forms. For a math problem, the model might:

manipulations)

- Derive the answer symbolically (substitutions, algebraic

- Derive it numerically (guess-and-check, tabulation)

- Derive it geometrically (a diagram, an area argument)

- Derive it by analogy to a solved instance

- Derive it by verification (checking the answer's properties)

Each is a distinct realisation of the same commitment (*valid derivation*). A verifier prompted with "*is this reasoning valid?*" will be trained on some surface distribution and will overfire or underfire on distributions it has not seen. Adversarial reasoning generation — producing plausible-looking but invalid reasoning that matches the verifier's surface distribution — is exactly what DR5 predicts is possible when the verifier is proposition-independent.

Where the placebo fails. A placebo of "obviously wrong reasoning" tests only the class of wrong-reasoning surface forms the placebo designer imagined. Adversarial wrong-reasoning that mimics the surface form of valid reasoning will pass the verifier. This is the same wall as the DCR1f Maxwell hit: the "placebo was invalid" reading (realisations of D we did not anticipate at the placebo cut) is formally indistinguishable from the "verifier is projecting" reading without an external g for "valid reasoning."

Practical guidance. If step-independence is a defining property of the verifier (as it typically is for cheap LLM judges), do not use its placebo-vs-positive delta as the correctness signal. Supply an independent structural check (g) — a proof checker, a symbolic executor, a constraint solver — and score the LLM against g's output rather than treating the LLM as g itself.

4. Corollary 3 — Retrieval-augmented factual grounding

Setting. An LLM generates a claim; a citation is attached; a verifier checks whether the citation supports the claim. D is "*the cited source establishes the claim.*"

Where D becomes multi-realisation. A citation "supports" a claim in many surface forms:

- Direct assertion: "*X is true.*"
- Definitional: the citation *defines* X such that the claim is trivial.
- Implicit: the citation asserts Y where Y presupposes X.
- Deductive: the citation asserts premises from which X follows.
- Statistical: the citation reports data consistent with X.

A verifier prompted with "*does the citation support the claim?*" will train on some subset of these surface forms and miss others. Both false positives (citation surface-matches but does not actually support) and false negatives (citation supports but not in a form the verifier recognises) follow directly from DR5.

Where the placebo fails. A placebo of "citation clearly unrelated to claim" tests only unrelated citations, not citations that surface-match but do not actually support. The DR5 wall predicts that adversarial retrieval — retrieving citations that surface-match the claim's vocabulary without supporting it — will pass proposition- independent verifiers.

Practical guidance. Do not treat citation quality as a proposition-level scoring problem. Impose a structural check: does the citation's content, under some canonical semantic form, entail the claim? This is exactly the g DR5 requires. Absent it, the verifier is above the wall only for the surface forms it happens to have seen.

5. Corollary 4 — Latent goal identification

Setting. An observer watches an agent take actions and tries to identify the agent's goal. D is the goal; each action a_i is a proposition-analogue; a verifier scores each action against D independently.

Where D becomes multi-realisation. A goal admits multiple action realisations trivially: any goal that can be achieved by more than one sequence of actions has multiple realisations. In practice:

there.

- Goal = "*navigate to the kitchen.*"
- Realisation 1: walk directly.
- Realisation 2: walk to the living room first, then kitchen.
- Realisation 3: pretend not to be going to the kitchen, then arrive

Realisation 3 is adversarial. A verifier that checks per-action plausibility against D will judge each of Realisation 3's actions individually plausible (or implausibly, depending on the surface form), and will not aggregate them into the goal-recognition question.

Where the placebo fails. A placebo of "agent doing something else" tests only obvious alternatives, not adversarial realisations of the target goal. DR5 predicts that any agent whose action distribution partially overlaps the goal's realisation set will fool a proposition- independent verifier.

Practical guidance. Goal recognition is inherently a class-level problem, not an action-level problem. Trying to build a per-action goal-detector without an explicit goal-representation g is above the DR5 wall only for goals with canonical action forms.

6. What DR5b does not claim

theorem; DR5b enumerates settings where the theorem operates. Each corollary is stated as an application, not as a separate theorem.

multi-realisation targets.** It claims that proposition-independent scoring is insufficient. Class-aware nomination (with a sound g) is the escape route DR5 identifies.

That is per-domain work. The best DR5 can offer is: know that you need g, and know what "sound and complete for D" means for your specific D.

attempted) would run a concrete controlled experiment in one of these domains and report where the wall appears.

- **It does not claim these corollaries are proved.** DR5 is a
- **It does not claim verification is impossible for
- **It does not identify the correct g for any specific domain.**
- **It does not attempt empirical demonstration.** DR6 (planned but not

7. Practical guidance for verifier designers

Six specific questions that determine whether your verification protocol is above or below the DR5 wall for your specific target D:

corpus?** If yes, DR5 applies. If no, standard placebo-based verification is above the wall.

depends only on p, not on other propositions or an external group structure.) If yes, DR5 applies to N.

explicit g, then your protocol is doing implicit grouping — somewhere your pipeline is deciding *which* propositions belong to the target class before scoring. Find that step; treat it as g; audit it for soundness and completeness.

placebo" from "the placebo is not clean for D"?** If not, DR5 §5 applies: the ambiguity is irreducible in your protocol.

Spencer's candidate-selection circularity at the class level; the wall re-appears one level up.

structural problem?** A calibration problem is fixable by tuning N; a structural problem requires importing g or restating the question. DCR1f showed how to distinguish: if narrowing N to fix one problem worsens another (precision-recall tradeoff over the held-out register), the wall is structural, not calibratable.

1. **Does D admit multiple non-equivalent surface forms in your
2. **Is your scoring function N proposition-independent?** ($N(p)$
3. **What is your grouping function g?** If you do not have an
4. **Does your placebo distinguish "your verifier projects on the
5. **Is g derived from the same signal as N?** If yes, you have
6. **Is the wall you are hitting a calibration problem or a

8. Where the arc goes from here

DR5 established the theorem. DR5b lists corollaries. Two named next-steps that would advance the framework meaningfully:

the four domains above (code correctness is the tractable candidate) and report whether the wall appears as DR5 predicts. This would move DR5 from "theorem with one worked instance" to "theorem with independent empirical corroboration."

constructed *soundly* for a specific D. What structural properties must the extractor have? Can g be learned? Under what conditions does the DCR1e-style "presupposition-inferring extractor implicitly defines g" observation generalise?

- **DR6 — empirical corollary.** Run a controlled experiment in one of
- **DR7 — grouping function correctness.** Study when g can be

Neither is trivial. Both are follow-ups DR5 opens rather than closes.

Appendix: reproduction

DR5b is a paper-only artifact. It does not add code or run experiments. The DCR arc's empirical corroboration reproduces via each paper's run_dcr1*.py and run_dcr2a.py. All prior verdicts unchanged.